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Studierichting: Informatica
Oefeningenreeks 3H nr.7.8

$r = \sin\theta + \cos\theta$

- domein van $r = \mathbb{R}$ en $p = 2\pi$
  - beperkte domein: $[0; 2\pi]$
- $r = 0 \Leftrightarrow \sin\theta + \cos\theta = 0$

  $\Leftrightarrow \dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\cos\theta} = 0$

  $\Leftrightarrow \tan\theta + 1 = 0$

  $\Leftrightarrow \tan\theta = -1 \Rightarrow \theta = \dfrac{3\pi}{4} + k\pi$

  - punten in domein: $\left\{\dfrac{3\pi}{4}, \dfrac{7\pi}{4}\right\}$
- $r' = 0 \Leftrightarrow \cos\theta - \sin\theta = 0$

  $\Leftrightarrow \dfrac{\cos\theta}{\cos\theta} - \dfrac{\sin\theta}{\cos\theta} = 0$

  $\Leftrightarrow 1 - \tan\theta = 0$

  $\Leftrightarrow \tan\theta = 1 \Rightarrow \theta = \dfrac{\pi}{4} + k\pi$

  - punten in domein: $\left\{\dfrac{\pi}{4}, \dfrac{5\pi}{4}\right\}$
- Tekenonderzoek:

θ	0		$\frac{\pi}{4}$		$\frac{3\pi}{4}$		$\frac{5\pi}{4}$		$\frac{7\pi}{4}$
r	$+$	$+$	$+$	$+$	0	$-$	$-$	$-$	0
r'	$-$	$-$	0	$+$	$+$	$+$	0	$-$	$-$
	$\searrow$	$\searrow$	$m(\sqrt{2})$	$\nearrow$	$\nearrow$	$\nearrow$	$M(-\sqrt{2})$	$\searrow$	$\searrow$

- Tekening:

References

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